



## HARD PLATE RF2

- Product name / model number: **DELTA V – PARA RF2**
- Armor type: HARD
- NIJ standard: 0101.07
- Claimed threat level: RF2
- Construction description:

## Materials and Manufacturing Description

### 1. Cutting of Ballistic Fabrics

The ballistic fabrics are cut using either an automatic CNC cutting machine or manual cutting methods. Cutting markers are generated based on size specifications and production quantities to optimize material yield.

### 2. Layer Stacking

After cutting, the ballistic fabrics are stacked according to size. In this configuration, 100 layers of UHMWPE fabric are used, each with an areal density of 130 g/m<sup>2</sup>.

### 3. Mold Loading

The stacked fabric layup is positioned inside the mold cavity of the thermoforming press.

### 4. Thermoforming and Consolidation

The thermoforming process is carried out under controlled pressure and temperature conditions. This thermo-mechanical cycle consolidates the fabric stack into a monolithic composite structure with a final thickness of 16 mm. The mold geometry defines the final plate shape while integrating all fabric layers into a single consolidated body.

### 5. Front Ceramic Strike Face Bonding

A 5 mm thick monolithic Silicon Carbide (SiC) ceramic plate is bonded to the front (strike face) of the consolidated UHMWPE backing using a contact adhesive system. The purpose of the ceramic strike face is to disrupt, blunt, and erode steel-core projectiles (e.g., M855 / SS109), thereby reducing penetrator effectiveness before energy absorption by the composite backing.

### 6. Aramid Spall Control Layer

Two woven aramid fabric layers are bonded to the Silicon Carbide plate using a





polyurethane adhesive. This layer functions as a spall mitigation and fragment containment system.

**7. Trauma Reduction Panel Integration**

A trauma reduction panel is cut and bonded to the rear face of each plate (the side in contact with the user). In this case, the trauma pad consists of cross-linked polyolefin foam with an areal density of 100 g/m<sup>2</sup>. The foam is bonded to the UHMWPE backing using contact adhesive.

**8. External Protective Encapsulation**

The fully assembled plate (SiC strike face + UHMWPE backing + trauma pad) is encapsulated with 1000 denier ripstop nylon fabric, pre-laminated with contact adhesive to ensure full surface bonding and environmental protection.

**9. Final Assembly**

A front and rear plate are inserted into the corresponding carrier system (plate carrier vest) to configure a ballistic vest providing front and back RF2 ballistic protection.

- Nominal dimensions:

| Size     | Width [mm] | Length [mm] |
|----------|------------|-------------|
| Small    | 222        | 300         |
| Medium   | 240        | 300         |
| Standard | 254        | 305         |
| Large    | 260        | 320         |

